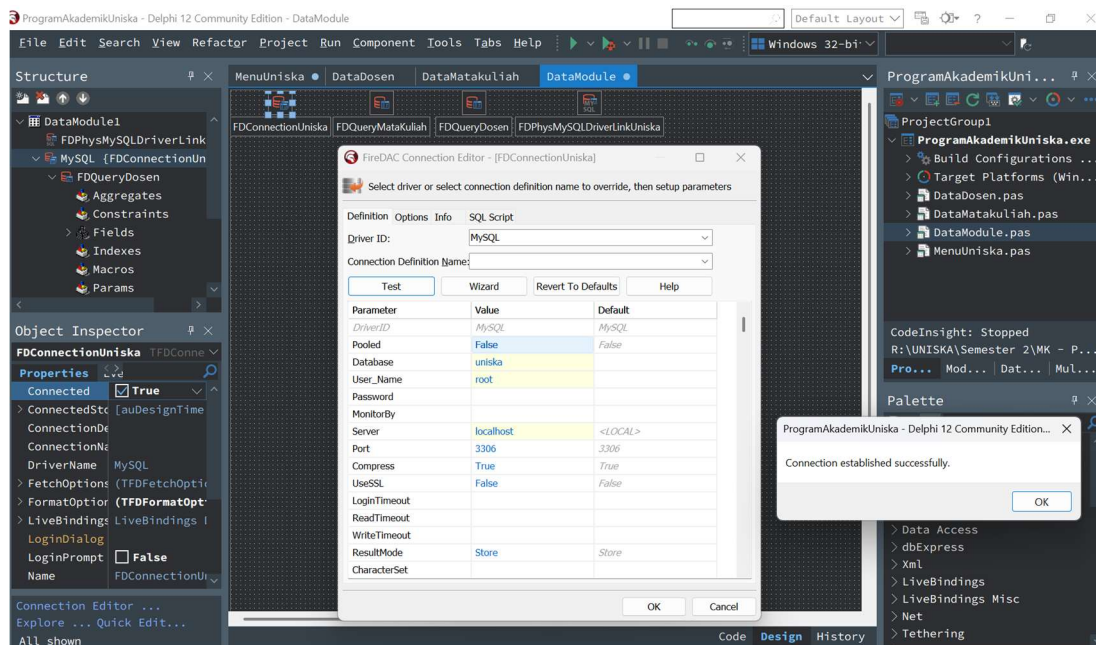


Menghubungkan Program Delphi dengan Database `uniska` menggunakan MySQL

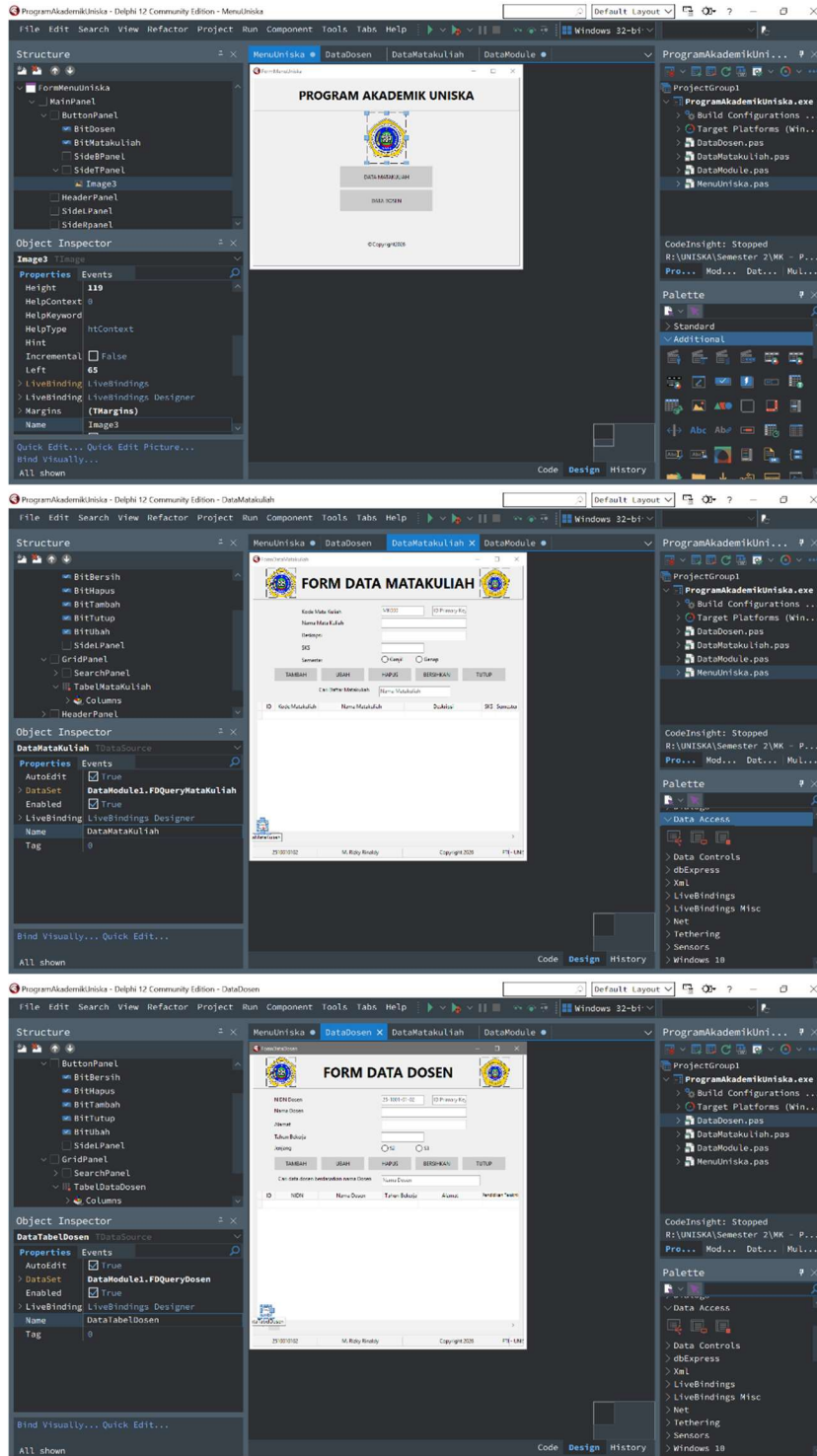
Dosen Pembimbing :
Muhammad Saidi Rahman, S.Kom, M.Kom



M. Rizky Rinaldy
2510010102

**FAKULTAS TEKNOLOGI INFORMASI
UNIVERSITAS ISLAM KALIMANTAN
MUHAMMAD ARSYAD AL-BANJARY**

DesainForm_ProgramAkademikUniska



SourceCode_MenuUniska.pas

```
1  unit MenuUniska;
2
3  interface
4
5  uses
6      Winapi.Windows, Winapi.Messages, System.SysUtils, System.Variants, System.Classes, Vcl.Graphics,
7      Vcl.Controls, Vcl.Forms, Vcl.Dialogs, Vcl.StdCtrls, Vcl.Buttons,
8      Vcl.Imaging.pngimage, Vcl.ExtCtrls, Vcl.ComCtrls,
9      DataModule, DataMataKuliah, DataDosen ;
10
11 type
12     TFormMenuUniska = class(TForm)
13     MainPanel: TPanel;
14     HeaderPanel: TPanel;
15     ButtonPanel: TPanel;
16     BitDosen: TBitBtn;
17     BitMatakuliah: TBitBtn;
18     SideLPanel: TPanel;
19     SideRpanel: TPanel;
20     SideTPanel: TPanel;
21     SideBPanel: TPanel;
22     Image3: TImage;
23     procedure BitMatakuliahClick(Sender: TObject);
24     procedure BitDosenClick(Sender: TObject);
25 private
26     { Private declarations }
27 public
28     { Public declarations }
29 end;
30
31 var
32     FormMenuUniska: TFormMenuUniska;
33
34 implementation
35
36 {$R *.dfm}
37
38 procedure TFormMenuUniska.BitDosenClick(Sender: TObject);
39 begin
40     FormDataDosen.Show
41 end;
42
43 procedure TFormMenuUniska.BitMatakuliahClick(Sender: TObject);
44 begin
45     FormMataKuliah.Show;
46 end;
47
48 end.
49
```

SourceCode_DataMatakuliah.pas

```

1  unit DataMatakuliah;
2
3  interface
4
5  uses
6  Winapi.Windows, Winapi.Messages, System.SysUtils, System.Variants, System.Classes, Vcl.Graphics,
7  Vcl.Controls, Vcl.Forms, Vcl.Dialogs, Vcl.StdCtrls, Vcl.ComCtrls, Vcl.ExtCtrls,
8  Data.DB, Vcl.Grids, Vcl.DBGrids,
9  DataModule, Vcl.Buttons, Vcl.Imaging.pngimage;
10
11 type
12   TFormMataKuliah = class(TForm)
13   MainPanel: TPanel;
14   StatusBarRizky: TStatusBar;
15   HeaderPanel: TPanel;
16   InputPanel: TPanel;
17   Label1: TLabel;
18   EdKodeMK: TEdit;
19   ButtonPanel: TPanel;
20   GridPanel: TPanel;
21   TabelMataKuliah: TDBGrid;
22   DataMataKuliah: TDataSource;
23   EdNamaMK: TEdit;
24   Label2: TLabel;
25   EdDeskripsi: TEdit;
26   Label3: TLabel;
27   EdSKS: TEdit;
28   Label4: TLabel;
29   Label5: TLabel;
30   RBGanjil: TRadioButton;
31   RBGenap: TRadioButton;
32   BitTutup: TBitBtn;
33   BitBersih: TBitBtn;
34   BitHapus: TBitBtn;
35   BitUbah: TBitBtn;
36   BitTambah: TBitBtn;
37   SidelPanel: TPanel;
38   Image1: TImage;
39   Image2: TImage;
40   SearchPanel: TPanel;
41   SearchBoxMK: TEdit;
42   Label6: TLabel;
43   EdID: TEdit;
44   procedure FormCreate(Sender: TObject);
45   procedure BitTambahClick(Sender: TObject);
46   procedure TabelMataKuliahCellClick(Column: TColumn);
47   procedure BitUbahClick(Sender: TObject);
48   procedure BitHapusClick(Sender: TObject);
49   procedure BitBersihClick(Sender: TObject);
50   procedure BitTutupClick(Sender: TObject);
51   procedure SearchBoxMKChange(Sender: TObject);
52 private
53   procedure bersih();
54   procedure tampil_data();
55   { Private declarations }
56 public
57   { Public declarations }
58 end;
59
60 var
61   FormMataKuliah: TFormMataKuliah;
62
63 implementation
64
65 {$R *.dfm}
66
67 { TFormMataKuliah }
68
69 procedure TFormMataKuliah.bersih;
70 begin
71   EdID.Clear;
72   EdKodeMK.Clear;
73   EdNamaMK.Clear;
74   EdDeskripsi.Clear;
75   EdSKS.Clear;
76   BitTambah.Enabled := True;
77   BitUbah.Enabled := False;
78   BitHapus.Enabled := False;
79 end;
80
81 procedure TFormMataKuliah.BitBersihClick(Sender: TObject);
82 begin
83   bersih;
84   DataModule1.FDQueryMataKuliah.Close;
85   DataModule1.FDQueryMataKuliah.sql.Text := 'SELECT * FROM mata_kuliah';
86   DataModule1.FDQueryMataKuliah.Open();
87 end;
88
89 procedure TFormMataKuliah.BitHapusClick(Sender: TObject);
90 var
91   IsiTabelKosong: Boolean;
92 begin
93   if (messagedlg('Hapus data ini?', mtconfirmation, [mbyes, mbno], 0) = mbyes) then
94     begin
95       IsiTabelKosong := DataModule1.FDQueryMataKuliah.RecordCount <= 1;
96       DataModule1.FDQueryMataKuliah.sql.Text := 'DELETE FROM mata_kuliah WHERE id=:id';
97       DataModule1.FDQueryMataKuliah.ParamByName('id').AsString := EdID.Text;
98       DataModule1.FDQueryMataKuliah.ExecSQL;
99       if IsiTabelKosong then
100         begin
101           DataModule1.FDQueryMataKuliah.sql.Text := 'ALTER TABLE mata_kuliah AUTO_INCREMENT = 1';
102           DataModule1.FDQueryMataKuliah.ExecSQL;
103         end;
104       Showmessage('Data Matakuliah berhasil dihapus');
105       bersih;
106       tampil_data;
107     end;
108 end;
109
110 end;
111
112 procedure TFormMataKuliah.BitTambahClick(Sender: TObject);
113 begin
114   if (EdKodeMK.Text = '') or (EdNamaMK.Text = '') or (EdDeskripsi.Text = '') or (EdSKS.Text = '') then
115     begin
116       messagedlg('Harap isi data dengan lengkap', mtwarning, [mbok], 0)
117     end
118   else
119     begin
120       DataModule1.FDQueryMataKuliah.sql.Text := 'INSERT INTO mata_kuliah VALUES (null, :kode, :nama, :desk, :sks, :semester)';
121       DataModule1.FDQueryMataKuliah.ParamByName('kode').AsString := EdKodeMK.Text;
122       DataModule1.FDQueryMataKuliah.ParamByName('nama').AsString := EdNamaMK.Text;
123       DataModule1.FDQueryMataKuliah.ParamByName('desk').AsString := EdDeskripsi.Text;
124       DataModule1.FDQueryMataKuliah.ParamByName('sks').AsString := EdSKS.Text;
125       if RBGanjil.Checked = True then
126         DataModule1.FDQueryMataKuliah.ParamByName('semester').AsString := 'Ganjil'
127       else
128         DataModule1.FDQueryMataKuliah.ParamByName('semester').AsString := 'Genap';
129       DataModule1.FDQueryMataKuliah.ExecSQL;
130       Showmessage('Data Matakuliah berhasil ditambahkan!');
131       bersih;
132     end;
133 end;
134
135 procedure TFormMataKuliah.BitTutupClick(Sender: TObject);
136 begin
137   close;
138 end;
139
140 procedure TFormMataKuliah.BitUbahClick(Sender: TObject);
141 begin
142   if (EdKodeMK.Text = '') or (EdNamaMK.Text = '') or (EdDeskripsi.Text = '') or (EdSKS.Text = '') then
143     begin
144       messagedlg('Harap isi data dengan lengkap', mtwarning, [mbok], 0)
145     end
146   else
147     begin
148       DataModule1.FDQueryMataKuliah.sql.Text := 'UPDATE mata_kuliah SET kode_mk=:kode, nama_mk=:nama, deskripsi=:desk, sks=:sks, semester=:semester WHERE id=:id';
149       DataModule1.FDQueryMataKuliah.ParamByName('kode').AsString := EdKodeMK.Text;
150       DataModule1.FDQueryMataKuliah.ParamByName('nama').AsString := EdNamaMK.Text;
151       DataModule1.FDQueryMataKuliah.ParamByName('desk').AsString := EdDeskripsi.Text;
152       DataModule1.FDQueryMataKuliah.ParamByName('sks').AsString := EdSKS.Text;
153       if RBGanjil.Checked = True then
154         DataModule1.FDQueryMataKuliah.ParamByName('sm').AsString := 'Ganjil'
155       else
156         DataModule1.FDQueryMataKuliah.ParamByName('sm').AsString := 'Genap';
157       DataModule1.FDQueryMataKuliah.ParamByName('id').AsString := EdID.Text;
158       DataModule1.FDQueryMataKuliah.ExecSQL;
159       Showmessage('Data Matakuliah berhasil diperbaharui!');
160       bersih;
161     end;
162 end;
163
164 procedure TFormMataKuliah.FormCreate(Sender: TObject);
165 begin
166   bersih;
167   (TabelMataKuliah.Columns[0].Width := 50;
168   TabelMataKuliah.Columns[1].Width := 70;
169   TabelMataKuliah.Columns[2].Width := 100;
170   TabelMataKuliah.Columns[3].Width := 200;
171 end;
172
173 procedure TFormMataKuliah.SearchBoxMKChange(Sender: TObject);
174 begin
175   DataModule1.FDQueryMataKuliah.Close;
176   DataModule1.FDQueryMataKuliah.sql.Text := 'SELECT * FROM mata_kuliah WHERE nama_mk like :nama';
177   DataModule1.FDQueryMataKuliah.ParamByName('nama').AsString := '%' + SearchBoxMK.Text + '%';
178   DataModule1.FDQueryMataKuliah.Open();
179 end;
180
181 procedure TFormMataKuliah.TabelMataKuliahCellClick(Column: TColumn);
182 begin
183   EdID.Text := TabelMataKuliah.Fields[0].AsString;
184   EdKodeMK.Text := TabelMataKuliah.Fields[1].AsString;
185   EdNamaMK.Text := TabelMataKuliah.Fields[2].AsString;
186   EdDeskripsi.Text := TabelMataKuliah.Fields[3].AsString;
187   EdSKS.Text := TabelMataKuliah.Fields[4].AsString;
188   if (TabelMataKuliah.Fields[5].AsString = 'Ganjil') then
189     RBGanjil.Checked := True
190   else
191     RBGenap.Checked := True;
192   BitTambah.Enabled := False;
193   BitUbah.Enabled := True;
194   BitHapus.Enabled := True;
195 end;
196
197 procedure TFormMataKuliah.tampil_data;
198 begin
199   DataModule1.FDQueryMataKuliah.Close;
200   DataModule1.FDQueryMataKuliah.sql.Text := 'SELECT * FROM mata_kuliah';
201   DataModule1.FDQueryMataKuliah.Open();
202 end;
203

```


SourceCode_DataDosen.pas

```

1 unit DataDosen;
2
3 interface
4
5 uses
6   Winapi.Windows, Winapi.Messages, System.SysUtils, System.Variants, System.Classes, Vcl.Graphics,
7   Vcl.Controls, Vcl.Forms, Vcl.Dialogs, Data.DB, Vcl.Grids, Vcl.DBGrids,
8   Vcl.StdCtrls, Vcl.Buttons, Vcl.Imaging.pngimage, Vcl.ExtCtrls, Vcl.ComCtrls, DataModule;
9
10 type
11   TFormDataDosen = class(TForm)
12   MainPanel: TPanel;
13   StatusBar1: TStatusBar;
14   HeaderPanel: TPanel;
15   Image1: TImage;
16   Image2: TImage;
17   InputPanel: TPanel;
18   Label1: TLabel;
19   Label2: TLabel;
20   Label3: TLabel;
21   Label4: TLabel;
22   EdNIDN: TEdit;
23   EdNamaDosen: TEdit;
24   EdAlamat: TEdit;
25   EdTahunBekerja: TEdit;
26   RBS2: TRadioButton;
27   RBS3: TRadioButton;
28   EdID: TEdit;
29   ButtonPanel: TPanel;
30   BitTutup: TBitBtn;
31   BitBersih: TBitBtn;
32   BitHapus: TBitBtn;
33   BitUbah: TBitBtn;
34   BitTambah: TBitBtn;
35   SidePanel: TPanel;
36   GridPanel: TPanel;
37   TabelDataDosen: TDBGrid;
38   SearchPanel: TPanel;
39   Label6: TLabel;
40   SearchBoxDosen: TEdit;
41   DataTabelDosen: TDataSource;
42   Label5: TLabel;
43   procedure FormCreate(Sender: TObject);
44   procedure BitBersihClick(Sender: TObject);
45   procedure BitTambahClick(Sender: TObject);
46   procedure BitHapusClick(Sender: TObject);
47   procedure BitUbahClick(Sender: TObject);
48   procedure BitTutupClick(Sender: TObject);
49   procedure SearchBoxDosenChange(Sender: TObject);
50   procedure TabelDataDosenCellClick(Column: TColumn);
51   private
52     procedure bersih();
53     procedure tampil_data();
54   { Private declarations }
55   public
56     { Public declarations }
57   end;
58
59 var
60   FormDataDosen: TFormDataDosen;
61
62 implementation
63
64 {$R *.dfm}
65
66 procedure TFormDataDosen.bersih;
67 begin
68   EdID.Clear;
69   EdNIDN.Clear;
70   EdNamaDosen.Clear;
71   EdAlamat.Clear;
72   EdTahunBekerja.Clear;
73   BitTambah.Enabled := True;
74   BitUbah.Enabled := False;
75   BitHapus.Enabled := False;
76 end;
77
78 procedure TFormDataDosen.BitBersihClick(Sender: TObject);
79 begin
80   bersih;
81   DataModule1.FDQueryDosen.Close;
82   DataModule1.FDQueryDosen.sql.Text := 'SELECT * FROM dosen';
83   DataModule1.FDQueryDosen.Open();
84 end;
85
86 procedure TFormDataDosen.BitHapusClick(Sender: TObject);
87 var
88   IsiTabelKosong: Boolean;
89 begin
90   if (messageDlg('Hapus data ini?', mtConfirmation, [mbYes, mbNo], 0) = mrYes) then
91   begin
92     IsiTabelKosong := DataModule1.FDQueryMatakuliah.RecordCount <= 1;
93     DataModule1.FDQueryDosen.sql.Text := 'DELETE FROM dosen WHERE id=:id';
94     DataModule1.FDQueryDosen.ParamByName('id').AsString := EdID.Text;
95     DataModule1.FDQueryDosen.ExecSQL;
96     if IsiTabelKosong then
97     begin
98       DataModule1.FDQueryDosen.sql.Text := 'ALTER TABLE dosen AUTO_INCREMENT = 1';
99       DataModule1.FDQueryDosen.ExecSQL;
100     end;
101     ShowMessage('Data Dosen berhasil dihapus');
102     bersih;
103     tampil_data;
104   end;
105 end;
106

```

```

94   DataModule1.FDQueryDosen.ParamByName('id').AsString := EdID.Text;
95   DataModule1.FDQueryDosen.ExecSQL;
96   if IsiTabelKosong then
97   begin
98     DataModule1.FDQueryDosen.sql.Text := 'ALTER TABLE dosen AUTO_INCREMENT = 1';
99     DataModule1.FDQueryDosen.ExecSQL;
100   end;
101   ShowMessage('Data Dosen berhasil dihapus');
102   bersih;
103   tampil_data;
104 end;
105 end;
106
107 procedure TFormDataDosen.BitTambahClick(Sender: TObject);
108 begin
109   if (EdNIDN.Text = '') or (EdNamaDosen.Text = '') or (EdAlamat.Text = '') or (EdTahunBekerja.Text = '') then
110   begin
111     messageDlg('Harap isi data dengan lengkap', mtWarning, [mbOk], 0)
112   end
113   else
114   begin
115     DataModule1.FDQueryDosen.sql.Text := 'INSERT dosen VALUES (null, :nidn, :nama, :tahun, :alamat, :jenjang)';
116     DataModule1.FDQueryDosen.ParamByName('nidn').AsString := EdNIDN.Text;
117     DataModule1.FDQueryDosen.ParamByName('nama').AsString := EdNamaDosen.Text;
118     DataModule1.FDQueryDosen.ParamByName('alamat').AsString := EdAlamat.Text;
119     DataModule1.FDQueryDosen.ParamByName('tahun').AsString := EdTahunBekerja.Text;
120     if RBS2.Checked = True then
121       DataModule1.FDQueryDosen.ParamByName('jenjang').AsString := 'S2'
122     else
123       DataModule1.FDQueryDosen.ParamByName('jenjang').AsString := 'S3';
124     DataModule1.FDQueryDosen.ExecSQL;
125     ShowMessage('Data Dosen berhasil ditambahkan!');
126     bersih;
127   end;
128 end;
129
130 procedure TFormDataDosen.BitTutupClick(Sender: TObject);
131 begin
132   Close;
133 end;
134
135 procedure TFormDataDosen.BitUbahClick(Sender: TObject);
136 begin
137   if (EdNIDN.Text = '') or (EdNamaDosen.Text = '') or (EdAlamat.Text = '') or (EdTahunBekerja.Text = '') then
138   begin
139     messageDlg('Harap isi data dengan lengkap', mtWarning, [mbOk], 0)
140   end
141   else
142   begin
143     DataModule1.FDQueryDosen.sql.Text := 'UPDATE dosen SET nidn=:nidn, nama_dosen=:nama, tahun_bekerja=:tahun, alamat=:alamat, pendidikan_terakhir=:jenjang';
144     DataModule1.FDQueryDosen.ParamByName('nidn').AsString := EdNIDN.Text;
145     DataModule1.FDQueryDosen.ParamByName('nama').AsString := EdNamaDosen.Text;
146     DataModule1.FDQueryDosen.ParamByName('alamat').AsString := EdAlamat.Text;
147     DataModule1.FDQueryDosen.ParamByName('tahun').AsString := EdTahunBekerja.Text;
148     if RBS2.Checked = True then
149       DataModule1.FDQueryDosen.ParamByName('jenjang').AsString := 'S2'
150     else
151       DataModule1.FDQueryDosen.ParamByName('jenjang').AsString := 'S3';
152     DataModule1.FDQueryDosen.ParamByName('id').AsString := EdID.Text;
153     DataModule1.FDQueryDosen.ExecSQL;
154     ShowMessage('Data Dosen berhasil diperbaharui!');
155     bersih;
156   end;
157 end;
158
159 procedure TFormDataDosen.FormCreate(Sender: TObject);
160 begin
161   bersih;
162   (TabelDataDosen.Columns[0].Width := 50;
163   TabelDataDosen.Columns[1].Width := 70;
164   TabelDataDosen.Columns[2].Width := 100;
165   TabelDataDosen.Columns[3].Width := 200;});
166 end;
167
168 procedure TFormDataDosen.SearchBoxDosenChange(Sender: TObject);
169 begin
170   DataModule1.FDQueryDosen.Close;
171   DataModule1.FDQueryDosen.sql.Text := 'SELECT * FROM dosen WHERE nama_dosen like :nama';
172   DataModule1.FDQueryDosen.ParamByName('nama').AsString := '%' + SearchBoxDosen.Text + '%';
173   DataModule1.FDQueryDosen.Open();
174 end;
175
176 procedure TFormDataDosen.TabelDataDosenCellClick(Column: TColumn);
177 begin
178   EdID.Text := TabelDataDosen.Fields[0].AsString;
179   EdNIDN.Text := TabelDataDosen.Fields[1].AsString;
180   EdNamaDosen.Text := TabelDataDosen.Fields[2].AsString;
181   EdTahunBekerja.Text := TabelDataDosen.Fields[3].AsString;
182   EdAlamat.Text := TabelDataDosen.Fields[4].AsString;
183   if (TabelDataDosen.Fields[5].AsString = 'S2') then
184     RBS2.Checked := True
185   else
186     RBS3.Checked := True;
187   BitTambah.Enabled := False;
188   BitUbah.Enabled := True;
189   BitHapus.Enabled := True;
190 end;
191
192 procedure TFormDataDosen.tampil_data;
193 begin
194   DataModule1.FDQueryDosen.Close;
195   DataModule1.FDQueryDosen.sql.Text := 'SELECT * FROM dosen';
196   DataModule1.FDQueryDosen.Open();
197 end;
198
199 end.
200

```

2510010102

M. Rizky Rinaldy

Tampilan_ProgramAkademikUniska

FormMenuUniska

PROGRAM AKADEMIK UNISKA



DATA MATAKULIAH

DATA DOSEN

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FormDataMatakuliah

FORM DATA MATAKULIAH

Kode Mata Kuliah: MK0016
Nama Mata Kuliah: Pemrograman Visual
Deskripsi: Perancangan Aplikasi Desktop berbasis
SKS: 3
Semester: ☐ Ganjil ☒ Genap

TAMBAH UBAH HAPUS BERSIHKAN TUTUP

Cari Daftar Matakuliah: Nama Matakuliah

ID	Kode Matakuliah	Nama Matakuliah	Deskripsi	SKS	Semester
4	MK004	Pemrograman Berorientasi Objek	Konsep OOP menggunakan bahasa	3	Geni
5	MK005	Sistem Operasi	Manajemen memori, proses, dan	3	Gan
6	MK006	Jaringan Komputer	Konsep dasar topologi, protokol,	3	Geni
7	MK007	Rekayasa Perangkat Lunak	Metodologi dan siklus hidup peng	3	Gan
8	MK008	Kecerdasan Buatan	Pengenalan AI, machine learning,	3	Geni
9	MK009	Keamanan Informasi	Prinsip kriptografi dan keamanan	2	Gan
10	MK010	Interaksi Manusia dan Komputer	Perancangan antarmuka pengguna	2	Geni
11	MK011	Arsitektur Komputer	Struktur internal dan organisasi pe	3	Gan
12	MK012	Pemrograman Web	Pengembangan aplikasi web berb	3	Geni
13	MK013	Analisis dan Desain Sistem	Analisis kebutuhan sistem inform	3	Gan
14	MK014	Grafika Komputer	Pemrosesan citra digital dan pemi	3	Geni
15	MK015	Sistem Pendukung Keputusan	Metode pengambilan keputusan l	2	Gan
17	MK016	Pemrograman Visual	Perancangan Aplikasi Desktop ber	3	Geni

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FormDataDosen

FORM DATA DOSEN

NIDN Dosen: 2510010102
Nama Dosen: M. Rizky Rinaldy
Alamat: Banjarbaru, Kalimantan Selatan
Tahun Bekerja: 2030
Jenjang: ☐ S2 ☒ S3

TAMBAH UBAH HAPUS BERSIHKAN TUTUP

Cari data dosen berdasarkan nama Dosen: Nama Dosen

ID	NIDN	Nama Dosen	Tahun Bekerja	Alamat	Pendidikan Terakhir
2	2510010102	M. Rizky Rinaldy	2030	Banjarbaru, Kalimantan Selat	S3

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